8.4: Enzymes speed up metabolic reactions by lowering energy barriers

The laws of thermodynamics tell us what will and will not happen under given conditions but say nothing about the rate of these processes. A spontaneous chemical reaction occurs without any requirement for outside energy, but it may occur so slowly that it is imperceptible. For example, even though the hydrolysis of sucrose (table sugar) to glucose and fructose is exergonic, occurring spontaneously with a release of free energy ($\Delta G = -7$ kcal/mol), a solution of sucrose dissolved in sterile water will sit for years at room temperature with no appreciable hydrolysis. However, if we add a small amount of the enzyme sucrase to the solution, then all the sucrose may be hydrolyzed within seconds, as shown here:

The reaction sucrose (C 12 H 22 O 11) + H 2 O is catalyzed by sucrase (an enzyme) to form Glucose (C 6 H 12 O 6) + Fructose (C 6 H 12 O 6).

Sucrose hydrolysis

How does the enzyme do this? An **enzyme** is a macromolecule that acts as a ****catalyst**, a chemical agent that speeds up a reaction without being consumed by the reaction. In this chapter, we focus on enzymes that are proteins. (Some RNA molecules, called ribozymes, can function as enzymes; these will be discussed in Concepts 17.3 and 25.1.) Without regulation by enzymes, chemical traffic through the pathways of metabolism would become terribly congested because many chemical reactions would take such a long time. In the next two sections, we will see why spontaneous reactions can be slow and how an enzyme changes the situation.

The Activation Energy Barrier

Every chemical reaction between molecules involves both bond breaking and bond forming. For example, the hydrolysis of sucrose involves breaking the bond between glucose and fructose and one of the bonds of a water molecule and then forming two new bonds, as shown previously. Changing one molecule into another generally involves contorting the starting molecule into a highly unstable state before the reaction can proceed. This contortion can be compared to the bending of a metal key ring when you pry it open to add a new key. The key ring is highly unstable in its opened form but returns to a stable state once the key is threaded all the way onto the ring. To reach the contorted state where bonds can change, reactant molecules must absorb energy from their surroundings. When the new bonds of the product molecules form, energy is released as heat, and the molecules return to stable shapes with lower energy than the contorted state.

The initial investment of energy for starting a reaction—the energy required to contort the reactant molecules so the bonds can break—is known as the *free energy of activation*, or **activation energy**, abbreviated EA in this book. We can think of activation energy as the amount of energy needed to push the reactants to the top of an energy barrier, or "uphill," so that the "downhill" part of the reaction can begin. Activation energy is often supplied by heat in the form of thermal energy that the reactant molecules absorb from the surroundings. The absorption of thermal energy accelerates the reactant molecules, so they collide more often and more forcefully. It also agitates the atoms within the molecules, making the breakage of bonds more likely. When the molecules have absorbed enough energy for the bonds to break, the reactants are in an unstable condition known as the *transition state*.

Figure 8.13 graphs the energy changes for a hypothetical exergonic reaction that swaps portions of two reactant molecules:

AB+CD→AC+BReactants Products

Figure 8.13

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Energy profile of an exergonic reaction.
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A graph plots Free energy versus Progress of the reaction.

Reactants A B and C D must absorb enough energy from the surroundings to reach the unstable transition state where bonds can break. After bonds break, new bonds form, releasing energy to the surroundings. Graph showing energy profile of an exergonic reaction. Free energy is on the y-axis and Progress of the reaction is on the x-axis. Reactants A B and C D start at an initial state; they then must absorb enough energy from the surroundings to reach the unstable transition state where bonds can break (E sub A). After bonds have broken, new bonds form, releasing energy to the surroundings. The products A C and B D are formed (delta G less than zero).

The "molecules" are hypothetical, with A, B, C, and D representing portions of the molecules. Thermodynamically, this is an exergonic reaction, with a negative ΔG, and the reaction occurs spontaneously. However, the activation energy (EA) provides a barrier that determines the rate of the reaction.

DRAW IT *Graph the progress of an endergonic reaction in which EF and GH form products EG and FH, assuming that the reactants must pass through a transition state.*

*For suggested answer, see*Appendix A*.*

The activation of the reactants is represented by the uphill portion of the graph, in which the free-energy content of the reactant molecules is increasing. At the summit, when energy equivalent to EA has been absorbed, the reactants are in the transition state: They are activated, and their bonds can be broken. As the atoms then settle into their new, more stable bonding arrangements, energy is released to the surroundings. This corresponds to the downhill part of the curve, which shows the loss of free energy by the molecules. The overall decrease in free energy means that EA is repaid with dividends, as the formation of new bonds releases more energy than was invested in the breaking of old bonds.

The reaction shown in Figure 8.13 is exergonic and occurs spontaneously ($\Delta G<0$). However, the activation energy provides a barrier that determines the rate of the reaction. The reactants must absorb enough energy to reach the top of the activation energy barrier before the reaction can occur. For some reactions, EA is modest enough that even at room temperature there is sufficient thermal energy for many of the reactant molecules to reach the transition state in a short time. In most cases, however, EA is so high and the transition state is reached so rarely that the reaction will hardly proceed at all. In these cases, the reaction will occur at a noticeable rate only if energy is provided, usually by heat. For example, the reaction of gasoline and oxygen is exergonic and will occur spontaneously, but energy is required for the molecules to reach the transition state and react. Only when the spark plugs fire in an automobile engine can there be the explosive release of energy that pushes the pistons. Without a spark, a mixture of gasoline hydrocarbons and oxygen will not react because the EA barrier is too high.

CheckPoint

Energy and Bonds in Chemical Reactions

Check your understanding before you move on, with the CheckPoint: Energy and Bonds in Chemical Reactions.

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How Enzymes Speed Up Reactions

Proteins, DNA, and other complex cellular molecules are rich in free energy and have the potential to decompose spontaneously; that is, the laws of thermodynamics favor their breakdown. These molecules only persist because at temperatures typical for cells, few molecules can make it over the hump of activation energy. The barriers for selected reactions must occasionally be surmounted, however, for cells to carry out the processes needed for life. Heat can increase the rate of a reaction by allowing reactants to attain the transition state more often, but this would not work well in biological systems. First, high temperature denatures proteins and kills cells. Second, heat would speed up *all* reactions, not just those that are needed.

Instead of heat, organisms carry out **catalysis**, the process by which a catalyst selectively speeds up a reaction without itself being consumed. (You learned about catalysts earlier in this section.)

An enzyme catalyzes a reaction by lowering the EA barrier **(Figure 8.14)**, enabling the reactant molecules to absorb enough energy to reach the transition state even at moderate temperatures, as we'll see shortly. It is crucial to note that *an enzyme cannot change the* ΔG*for a reaction; it cannot make an endergonic reaction exergonic*. Enzymes can only hasten reactions that would eventually occur anyway, but this enables the cell to have a dynamic metabolism, routing chemicals smoothly through metabolic pathways. Also, enzymes are very specific for the reactions they catalyze, so they determine which chemical processes will be going on in the cell at any given time.

Figure 8.14

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The effect of an enzyme on activation energy.
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Graph showing course of reaction with and without enzyme.

E sub A without enzyme is shown to be approximately twice as large as E sub A with enzyme. Delta G is unaffected by enzyme. Free energy is on the y-axis and Progress of the reaction is on the x-axis. E sub A without enzyme is shown to be approximately twice as large as E sub A with enzyme. Delta G is unaffected by enzyme.

Without affecting the free-energy change (ΔG) for a reaction, an enzyme speeds the reaction by reducing its activation energy (EA).

Animation

How Enzymes Work

image

image

Substrate Specificity of Enzymes

The reactant an enzyme acts on is referred to as the enzyme's **substrate**. The enzyme binds to its substrate (or substrates, when there are two or more reactants), forming an **enzyme-substrate complex**. While enzyme and substrate are joined, the catalytic action of the enzyme converts the substrate to the product (or products) of the reaction. The overall process can be summarized as follows:

Enzyme +Enzyme-Enzyme +Substrate(s)⇌substrate ⇌Product(s)complex

Most enzyme names end in *-ase*. (See if you can find three examples of enzymes in the lower left of Figure 6.32.) For example, the enzyme sucrase catalyzes the hydrolysis of the disaccharide sucrose into its two monosaccharides, glucose and fructose (see the diagram at the beginning of Concept 8.4):

Sucrase +Sucrose +H2O⇌Sucrase-sucrose-H2Ocomplex⇌Sucrase +Glucose +Fructose

The reaction catalyzed by each enzyme is very specific; an enzyme can recognize its specific substrate even among closely related compounds. For instance, sucrase will act only on sucrose and will not bind to other disaccharides, such as maltose. What accounts for this molecular recognition? Recall that most enzymes are proteins, and that proteins are macromolecules with unique 3-D configurations. The specificity of an enzyme results from its shape, which is a consequence of its amino acid sequence.

Only a restricted region of the enzyme molecule actually binds to the substrate. This region, called the **active site**, is typically a pocket or groove on the surface of the enzyme where catalysis occurs (**Figure 8.15a**; see also Figure 5.16). Usually, the active site is formed by only a few of the enzyme's amino acids, with the rest of the protein molecule providing a framework that determines the shape of the active site. The specificity of an enzyme is attributed to a complementary fit between the shape of its active site and the shape of the substrate.

Figure 8.15

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Induced fit between an enzyme and its substrate.
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Induced fit between an enzyme and its substrate.

Part a. In this space-filling model of the enzyme hexokinase (blue), the active site forms a groove on the surface. The enzyme's substrate is glucose (red).

Part b. When the substrate enters the active site, it forms weak bonds with the enzyme, inducing a change in the shape of the protein. This change allows additional weak bonds to form, causing the active site to enfold the substrate and hold it in place.

An enzyme is not a stiff structure locked into a given shape. In fact, recent work by biochemists has shown that enzymes (and other proteins) seem to "dance" between subtly different shapes in a dynamic equilibrium, with slight differences in free energy for each "pose." The shape that best fits the substrate isn't necessarily the one with the lowest energy, but during the very short time the enzyme takes on this shape, its active site can bind to the substrate. The active site itself is also not a rigid receptacle for the substrate. As shown in **Figure 8.15b**, when the substrate enters the active site, the enzyme changes shape slightly due to interactions between the substrate's chemical groups and chemical groups on the side chains of the amino acids that form the active site. This shape change makes the active site fit even more snugly around the

substrate. The tightening of the binding after initial contact—called **induced fit**—is like a clasping handshake. Induced fit brings chemical groups of the active site into positions that enhance their ability to catalyze the chemical reaction.

Catalysis in the Enzyme's Active Site

In most enzymatic reactions, the substrate is held in the active site by so-called weak interactions, such as hydrogen bonds and ionic bonds. The R groups of a few of the amino acids that make up the active site catalyze the conversion of substrate to product, and the product departs from the active site. The enzyme is then free to take another substrate molecule into its active site. The entire cycle happens so fast that a single enzyme molecule typically acts on about 1,000 substrate molecules per second, and some enzymes are even faster. Enzymes, like other catalysts, emerge from the reaction in their original form. Therefore, very small amounts of enzyme can have a huge metabolic impact by functioning over and over again in catalytic cycles. **Figure 8.16** shows a catalytic cycle involving two substrates and two products.

Figure 8.16

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The active site and catalytic cycle of an enzyme.
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The active site and catalytic cycle of an enzyme.

An enzyme can convert one or more reactant molecules to one or more product molecules. The enzyme shown here converts two substrate molecules to two product molecules.

Six steps are given.

- Substrates enter active site; enzyme changes shape such that its active site enfolds the substrates (induced fit);
- In the enzyme-substrate complex, substrates are held in the active site by weak interactions, such as hydrogen bonds and ionic bonds.
- The active site lowers E sub A and speeds up the reaction (see description in text).
- Substrates are converted to products.
- Products are released.
- Enzyme remains and active site is available for two new substrate molecules.

VISUAL SKILLS *The enzyme-substrate complex passes through a transition state (see*Figure 8.13*).* Label the part of the cycle where the transition state occurs.*

*For suggested answer, see*Appendix A.**

Animation

Enzymes: Steps in a Reaction

image

Most metabolic reactions are reversible, and an enzyme can catalyze either the forward or the reverse reaction, depending on which direction has a negative ΔG. This in turn depends mainly on the relative concentrations of reactants and products. The net effect is always in the direction of equilibrium.

Enzymes use a variety of mechanisms that lower activation energy and speed up a reaction (see Figure 8.16, **STEP 3**):

- In reactions involving two or more reactants, the active site provides a template on which the substrates can come together in the proper orientation for a reaction to occur between them.
- As the active site of an enzyme clutches the bound substrates, the enzyme may stretch the substrate molecules toward their transition state form, stressing and bending critical chemical bonds to be broken during the reaction. Because EA is proportional to the difficulty of breaking the bonds, distorting the substrate helps it approach the transition state and thus reduces the amount of free energy that must be absorbed to achieve that state.
- The active site may also provide a microenvironment that is more conducive to a certain reaction than the solution itself would be without the enzyme. For example, if the active site has amino acids with acidic R groups, it may provide a pocket of low pH in an otherwise neutral cell. Here, an acidic amino acid may facilitate H+ transfer to the substrate as a key step in catalyzing the reaction.
- Amino acids in the active site may directly participate in the chemical reaction. Sometimes this process involves brief covalent bonding between the substrate and the side chain of an amino acid of the enzyme. Subsequent steps restore the side chains to their original states so that the active site is the same after the reaction as it was before.

The rate at which a particular amount of enzyme converts substrate to product is partly a function of the initial concentration of the substrate: The more substrate molecules that are available, the more frequently they access the active sites of the enzyme molecules. However, there is a limit to how fast the reaction can be pushed by adding more substrate to a fixed concentration of enzyme. At some point, the concentration of substrate will be high enough that all enzyme molecules will have their active sites engaged. As soon as the product exits an active site, another substrate molecule enters. At this substrate concentration, the enzyme is said to be *saturated*, and the rate of the reaction is determined by the speed at which the active site converts

substrate to product. When an enzyme population is saturated, the only way to increase the rate of product formation is to add more enzyme. Cells often increase the rate of a reaction by producing more enzyme molecules. You can graph the overall progress of an enzymatic reaction in the **Scientific Skills Exercise**.

Scientific Skills Exercise

Making a Line Graph and Calculating a Slope

AP Science Practices 1, 3, 4, 5

Does the Rate of Glucose 6-Phosphatase Activity Change over Time in Isolated Liver Cells? Glucose 6-phosphatase, which is found in mammalian liver cells, is a key enzyme in control of blood glucose level. The enzyme catalyzes the breakdown of glucose 6-phosphate into glucose and inorganic phosphate (Ⓟi). These products are transported out of liver cells into the blood, increasing the blood glucose level. In this exercise, you will graph data from a time-course experiment that measured Ⓟi concentration in the buffer outside isolated liver cells, thus indirectly measuring glucose 6-phosphatase activity inside the cells.

How the Experiment Was Done Isolated rat liver cells were placed in a dish with buffer at physiological conditions (pH7.4,37°C). Glucose 6-phosphate (the substrate) was added so it could be taken up by cells and broken down by glucose 6-phosphatase. A sample of buffer was removed every 5minutes and the concentration of Ⓟi that had been transported out of the cells was determined.

Data from the Experiment

Table showing the data from an experiment showing changes in inorganic phosphate over time.

Data from S. R. Commerford et al., Diets enriched in sucrose or fat increase gluconeogenesis and G-6-Pase but not basal glucose production in rats, *American Journal of Physiology-Endocrinology and Metabolism* 283:E545–E555 (2002).

The table is as follows:

Time (min)	Concentration of P 1 (mu mol per m L)
0	0
5	10
10	90

15	180
20	270
25	330
30	355
35	355
40	355

INTERPRET THE DATA

- To see patterns in the data from a time-course experiment like this, it is helpful to graph the data. First, determine which set of data goes on each axis.
- What did the researchers intentionally vary in the experiment? This is the independent variable, which goes on the x-axis.
- What are the units (abbreviated) for the independent variable? Explain in words what the abbreviation stands for.
- What was measured by the researchers? This is the dependent variable, which goes on the y-axis.
- What does the units abbreviation stand for? Label each axis, including the units.
- Next, mark off the axes with just enough evenly spaced tick marks to accommodate the full set of data. Determine the span of data values for each axis.
- What is the largest value to go on the x-axis? What is a reasonable spacing for the tick marks, and what should be the highest one?
- What is the largest value to go on the y-axis? What is a reasonable spacing for the tick marks, and what should be the highest one?
- Plot the data points on your graph. Match each x-value with its partner y-value and place a point on the graph at that coordinate. Draw a line that connects the points. (For additional information about graphs, see the Scientific Skills Review in Appendix D.)
- Examine your graph and look for patterns in the data.
- Does the concentration of Ⓟi increase evenly throughout the course of the experiment? To answer this question, describe the pattern you see in the graph.

- What part of the graph shows the highest rate of enzyme activity? Consider that the rate of enzyme activity is related to the slope of the line, $\Delta y/\Delta x$ (the "rise" over the "run"), in μmol/(mL · min), with the steepest slope indicating the highest rate of enzyme activity. Calculate the rate of enzyme activity (slope) where the graph is steepest.
- Can you think of a biological explanation for the pattern you see?
- If your blood glucose level is low from skipping lunch, what reaction (discussed in this exercise) will occur in your liver cells? Write out the reaction and put the name of the enzyme over the reaction arrow. How will this reaction affect your blood glucose level?

A tired student reads.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Effects of Local Conditions on Enzyme Activity

The activity of an enzyme—how efficiently the enzyme functions—is affected by general environmental factors, such as temperature and pH. It can also be affected by chemicals that specifically influence that enzyme. In fact, researchers have learned much about enzyme function by employing such chemicals.

Effects of Temperature and pH

The 3-D structures of proteins are sensitive to their environment (see Figure 5.20). As a consequence, each enzyme works better under some conditions than others, because these *optimal conditions* favor the most active shape for the enzyme.

Temperature and pH are environmental factors important in the activity of an enzyme. Up to a point, the rate of an enzymatic reaction increases with increasing temperature, partly because substrates collide with active sites more frequently when the molecules move rapidly. Above that temperature, however, the speed of the enzymatic reaction drops sharply. The thermal agitation of the enzyme molecule disrupts the hydrogen bonds, ionic bonds, and other weak interactions that stabilize the active shape of the enzyme, and the protein molecule eventually denatures. Each enzyme has an optimal temperature at which its reaction rate is greatest. Without denaturing the enzyme, this temperature allows the greatest number of molecular collisions and the fastest conversion of the reactants to product molecules. Most human enzymes have optimal temperatures of about 35–40°C (close to human body temperature). The thermophilic bacteria that live in hot springs contain enzymes with optimal temperatures of 70°C or higher **(Figure 8.17a)**.

Figure 8.17

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Environmental factors affecting enzyme activity.
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Environmental factors affecting enzyme activity.

Part a. Two curves showing that the rate of an enzymatic reaction increases with increasing temperature up to the optimal temperature for the enzyme. Above that temperature the speed of the enzymatic reaction drops sharply. The photo shows thermophilic cyanobacteria (green) thriving in the hot water of a Nevada geyser. The graph compares the optimal temperatures for an enzyme from the thermophilic bacterium Thermus oshimai (75 degrees Celsius) and human enzymes (body temperature, 37 degrees Celsius). Two curves showing that the rate of an enzymatic reaction increase with increasing temperature up to the optimal temperature for the enzyme. Above that temperature the speed of the enzymatic reaction drops sharply. The rate of reaction is on the y-axis and the temperature (Celsius) from zero to 120 is on the x-axis. The optimal temperature for a typical human enzyme is 37 degrees Celsius with the curve starting at zero and ending at 50 degrees Celsius. The optimal temperature for thermophilic (heat-tolerant) bacteria is 77 degrees Celsius with the curve starting at 40 and ending at 90 degrees Celsius.

Part b. This graph shows the rate of reaction for two digestive enzymes (pepsin and trypsin) over a range of p H values. Optimal pH for two enzymes shown as bell shaped curves where the rate of reaction increases with increasing p H up to a point and then decreases. Optimal p H for pepsin (stomach enzyme) is approximately 2, with the curve starting at 0 and ending at 4. Optimal p H for trypsin (intestinal enzyme) is approximately 8, with the cure starting at 6 and ending at 10.

Each enzyme has an optimal **(a)** temperature and **(b)** pH that favor the most active shape of the protein molecule.

INTERPRET THE DATA *Looking at the graph in (b), what is the optimal* pH *for pepsin activity? Explain why natural selection might have resulted in the optimal* pH *for pepsin, a stomach enzyme (see*Figure 3.11*).* What is the optimal *pH* for trypsin?*

Just as each enzyme has an optimal temperature, it also has a pH at which it is most active. The optimal pH values for most enzymes fall in the range of pH 6–8, but there are exceptions. For example, pepsin, a digestive enzyme in the human stomach, works best at a very low pH. Such an acidic environment denatures most enzymes, but pepsin is well-adapted evolutionarily to maintain its functional 3-D structure in the acidic environment of the stomach. In contrast, trypsin, a digestive enzyme residing in the more alkaline environment of the human intestine would be denatured in the stomach **(Figure 8.17b)**.

Cofactors

Many enzymes require nonprotein helpers for catalytic activity, often for chemical processes like electron transfers that cannot easily be carried out by the amino acids in proteins. These adjuncts, called **cofactors**, may be bound tightly to the enzyme as permanent residents, or they may bind loosely and reversibly along with the substrate. The cofactors of some enzymes are inorganic, such as the metal atoms zinc, iron, and copper in ionic form. If the cofactor is an organic molecule, it is referred to, more specifically, as a **coenzyme**. Most vitamins are important in nutrition because they act as coenzymes or raw materials from which coenzymes are made.

Enzyme Inhibitors

Certain chemicals selectively inhibit the action of specific enzymes. Sometimes the inhibitor attaches to the enzyme by covalent bonds, in which case the inhibition is usually irreversible. Many enzyme inhibitors, however, bind to the enzyme by weak interactions, and when this occurs the inhibition is reversible. Some reversible inhibitors resemble the normal substrate molecule and compete for admission into the active site **(Figure 8.18a and b)**. These mimics, called **competitive inhibitors**, reduce the productivity of enzymes by blocking substrates from entering active sites. This kind of inhibition can be overcome by increasing the concentration of substrate so that as active sites become available, more substrate molecules than inhibitor molecules are around to gain entry to the sites.

Figure 8.18

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Inhibition of enzyme activity.
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Inhibition of enzyme activity.

Part a. Normal binding: a substrate can bind normally to the active site of an enzyme. Enzyme shown with an active site that the substrate fits into.

Part b. Competitive inhibition: A competitive inhibitor mimics the substrate, competing for the active site so that the substrate can't physically bind with the enzyme.

Part c. Noncompetitive inhibitor binds to the enzyme away from the active site, altering the shape of the enzyme so that even if substrate can bind, the active site functions less effectively.

Animation

Enzymes: Competitive Inhibition

image

**

Animation

Enzymes: Noncompetitive Inhibition

**

In contrast, **noncompetitive inhibitors** do not directly compete with the substrate to bind to the enzyme at the active site **(Figure 8.18c)**. Instead, they impede enzymatic reactions by binding to another part of the enzyme. This interaction causes the enzyme molecule to change its shape in such a way that the active site becomes much less effective at catalyzing the conversion of substrate to product.

Toxins and poisons are often irreversible enzyme inhibitors. An example is sarin, a nerve gas. In 2017, sarin was used in a chemical attack in Syria, killing about 100 people and injuring hundreds more. This small molecule binds covalently to the R group on the amino acid serine, which is found in the active site of acetylcholinesterase, an enzyme important in the nervous system. Other examples include the pesticides DDT and parathion, inhibitors of key enzymes in the nervous system. Finally, many antibiotics are inhibitors of specific enzymes in bacteria. For instance, penicillin blocks the active site of an enzyme that many bacteria use to make cell walls.

Citing enzyme inhibitors that are metabolic poisons may give the impression that enzyme inhibition is generally abnormal and harmful. In fact, molecules naturally present in the cell often regulate enzyme activity by acting as inhibitors. Such regulation—selective inhibition—is essential to the control of cellular metabolism, as you'll see in Concept 8.5.

The Evolution of Enzymes

EVOLUTION Thus far, biochemists have identified more than 4,000 different enzymes in various species, most likely a very small fraction of all enzymes. How did this grand profusion of enzymes arise? Recall that most enzymes are proteins, and proteins are encoded by genes. A permanent change in a gene, known as a *mutation*, can result in a protein with one or more changed amino acids. In the case of an enzyme, if the changed amino acids are in the active site or some other crucial region, the altered enzyme might have a novel activity or might bind to a different substrate. Under environmental conditions where the new function benefits the organism, natural selection would tend to favor the mutated form of the gene, causing it to persist in the population. This simplified model is generally accepted as the main way in which the multitude of different enzymes arose over the past few billion years of life's history. Data supporting this model have been collected by researchers using a lab procedure that mimics evolution in natural populations **(Figure 8.19)**.

Figure 8.19

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Mimicking evolution of an enzyme with a new function.
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Mimicking evolution of an enzyme with a new function.

Active site shown with two changed amino acids in the active site, two changed amino acids on the surface, and two changed amino acids near the active site.

Researchers tested whether the function of an enzyme called β-galactosidase, which breaks down the sugar lactose, could change over time in populations of the bacterium *Escherichia coli*. After seven rounds of mutation and selection in the lab, β-galactosidase, evolved into an enzyme specialized for breaking down a sugar other than lactose. This ribbon model shows one subunit of the altered enzyme; six amino acids (blue dots) were different.